

Technical Progress Report

Project name:	Biped robot		
Group member:	Yangnan.yan 2363610	Group number:	1
	Jiarui.li 2360655		
	xinhang.li 2363444		
	Jian.wang 2362488		
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Project manager:	Dr. Ki-Young Song		

1. Executive Summary

This report documents the technical progress of the biped robot project in the current week. The team has completed structural simplification, stability optimization, foot structure improvement, hardware compartment protection enhancement, large-scale vision dataset collection, and model training. The original robot structure required excessive motor torque that could not be met by the existing medium-size motors. To solve this issue, the team reduced the degrees of freedom to two per leg, shortened the leg length, lowered mechanical load, and significantly improved standing and moving stability. In addition, the foot structure was optimized to enhance ground contact and balance capability. The vision dataset was expanded to 1800 images, which greatly improved recognition accuracy and robustness. Initial tests show that the robot achieves much better stability, controllability, and motion feasibility. Remaining issues include minor joint jitter and occasional misidentification at extreme angles. The team will continue targeted optimization to realize stable, continuous, and visually guided walking in the next stage.

2.Objectives & Scope

- Reduce the degrees of freedom to two to lower control complexity and motor load, and improve overall stability.
- Shorten leg length to match the driving capacity of existing medium-size motors.
- Optimize the foot structure to improve ground adaptability, friction, and standing balance.
- Expand the vision dataset to 1800 images and retrain the model for higher recognition success rate.
- Verify motion stability, balance performance, and vision recognition accuracy under the updated structure.
- Identify remaining issues in rigidity, smoothness, and robustness for subsequent improvement.
- Lay a solid foundation for stable walking and integrated vision-motion control.

3. Technical Work Completed

3.1 Structural Simplification and Stability Enhancement

The original structure required excessive motor power that exceeded the limit of medium-size motors. The team reduced the degrees of freedom to two, shortened leg length, simplified the actuation mechanism, reduced moment load, and significantly improved mechanical stability and motion feasibility. The adjusted structure was reassembled and verified.

3.2 Foot Structure Optimization

Redesigned and optimized the foot structure to increase contact area, improve friction and ground adaptability, enhance support stability, and reduce tipping risk during standing and stepping.

3.3 Hardware Protection and Component Reinforcement

Further improved the hardware compartment to reduce exposure of the ESP32, camera, battery, and wires. Key joints and transmission parts were reinforced to improve rigidity and anti-vibration performance.

3.4 Large-Scale Dataset Collection and Model Training

Collected and sorted a high-quality vision dataset of 1800 images with rich angles, distances, and lighting conditions. The model was retrained and deployed, which significantly improved recognition success rate and generalization ability.

3.5 System Integration and Preliminary Motion Test

Conducted power-on tests and basic walking attempts with the updated structure. Verified stability, balance performance, actuation smoothness, and real-time vision recognition under actual working conditions.

4. Observations & Findings

- Reducing degrees of freedom to two effectively lowered motor load, simplified control, and greatly improved standing and moving stability.
- The optimized foot structure significantly improved support balance and ground adaptability.
- Shortened leg length made the actuation system fully compatible with medium-size motors.
- With 1800 training images, the vision system achieved much higher recognition accuracy and robustness.
- Minor jitter exists in individual joints due to residual structural clearance.
- Occasional misidentification still occurs at extreme viewing angles but with much lower frequency.

5. Challenges & Roadblocks

- Although stability is greatly improved, residual joint clearance still causes minor jitter during movement.
- Extreme angles still lead to occasional recognition errors.
- Gait coordination and actuation timing require further fine-tuning for smoother walking.
- The trade-off between low degrees of freedom, motion flexibility, and stability needs continued optimization.

6. Next Steps & Recommendations

- Further eliminate joint clearance and strengthen fasteners to reduce jitter and improve rigidity.
- Fine-tune gait parameters and actuation timing to achieve smoother and more coordinated stepping.
- Supplement a small number of extreme-angle samples and fine-tune the model to further reduce errors.
- Fully enclose the hardware compartment for all-round protection.
- Conduct integrated tests of stable walking and real-time vision feedback to verify overall system performance.

7. Conclusion

In the current week, the team solved the core problem of excessive motor demand by reducing the degrees of freedom to two and shortening leg length, which significantly improved stability and made motion feasible with existing medium-size motors. The optimized foot structure further enhanced support balance and anti-tipping performance. Meanwhile, the vision model trained with 1800 images achieved greatly improved recognition success rate and stability. The updated mechanical structure and vision system have passed initial verification, and the overall performance has been significantly improved. Remaining issues include minor joint jitter and occasional



misidentification at extreme angles. In the next stage, the team will focus on joint optimization, gait refinement, model fine-tuning, and full protection design, so as to realize stable continuous walking and high-precision visual recognition for the biped robot.